

1 - PURPOSE

Lipid matrix tablets provide an attractive and practical approach to sustain the release of highly water soluble drugs. Lipid matrices are non-swellable and are insensitive to environmental and physiological conditions such as humidity or pH changes. Therefore, a tablet film coating is not required. However, the strength or robustness of a lipid matrix is dependent on both formulation and processing parameters. Sintering is a post-compression heat treatment frequently described in the literature as an excellent procedure to reinforce lipid matrix tablets, however the technique is not used industrially due to the processing conditions (80°C). This study demonstrates how to obtain the same benefits as sintering using standard curing conditions; the formulation utilizes Compritol® 888 ATO as the matrix former by direct compression and Precirol® ATO 5 as the matrix reinforcer by curing.

2 - INTRODUCTION

Compritol® 888 ATO is a lipid excipient frequently used in sustained release applications. It is composed of glycerol behenate exhibiting a HLB of 1 and a melting point of 70°C. Due to its lipophilic nature and reduced wettability, drug delivery can be successfully extended over prolonged periods of time and might be highly effective even for freely water soluble drugs.

Compritol® is not soluble in alcohol and therefore the lipid matrix structure produced by direct compression is not dissolvable in alcohol. This means the matrix is not subject to hydro-alcoholic media. However if the API has a higher solubility in hydro-alcoholic media, direct compression produces a lipid matrix with limited robustness because the mechanism of diffusion will be faster due to higher driving force.

However, it has been shown that a post-heat treatment on tablets or a hot process can generate a lipid belt that enhances the alcohol resistance. Sintering technique has been studied with Compritol® 888 when tablet obtained by direct compression was placed in a dry oven to 80°C. This extreme condition is not well accepted by industrials and then alternative lipids with lower melting point might be more adapted. Interestingly, Precirol® ATO 5 has a lower melting point 54°C and can be associated with Compritol® 888 ATO in tablet. A simple curing step at 60°C can melt Precirol® ATO 5 within the tablet.

3 - EXPERIMENTAL METHODS

Theophylline anhydrous was selected for this study because it is a water soluble drug (Table 1) and it is available in sustained release dosage forms. It is an interesting model for alcohol dose dumping since its solubility is enhanced by a factor of three in 0.1 N HCl with 40% ethanol (Table 2).

Table 1: Theophylline description.

Therapeutic Class	Bronchodilator
Water solubility at 25°C	1-5 mg/mL
Half-life	8 hours
Dose	100 - 600 mg
Melting point	272°C
pKa	8.81, weak base

Compritol® (glyceryl behenate - USP/NF) is used in the formulation as matrix former and Precirol® (glyceryl distearate - USP/NF) as a matrix reinforcer. A blend of diluents were added in order to get a conform hardness and weight in the final tablet (Table 3).

Table 3: Tablet formulation.

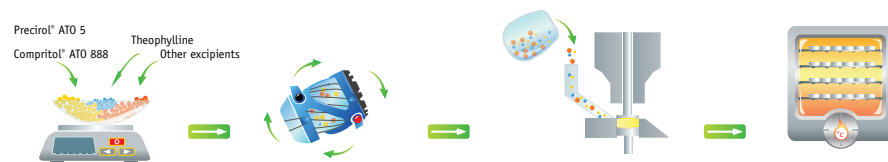
Ingredients	Supplier	mg/tablet	% w/w
Theophylline anhydrous	Sigma-Aldrich	100	16.7
Compritol® 888 ATO (glyceryl behenate)	Gattefossé	90	15.0
Precirol® ATO 5 (glyceryl distearate)	Gattefossé	60	10.0
Fujicalin SG (dibasic calcium phosphate anhydrous)	Fuji Chemical	217	36.2
Flowlac 100 (spray-dried lactose)	Meggles Pharma	109	18.1
Neusilin US2 (Mg-Al-metasilicate)	Fuji Chemical	18	3.0
Mg-stearate	Sigma-Aldrich	6	1.0
Total tablet weight		600 mg	100%

Lipid matrix tablets were prepared by direct compression using theophylline anhydrous, diluents and a blend of Compritol® 888 ATO and Precirol® ATO 5. All tablets were prepared on an alternative tableting machine (Korsch EK03). The contact angle on the tablet surface was measured using a goniometer ILMS from GBX instruments. A 3 µl droplet of demineralized water was used and pictures were taken during 90 s.

Tablets were then exposed to a curing at 60°C for 30 min in a dry oven. Dissolution studies were conducted in 0.1 N HCl (with or without 40% ethanol) and in phosphate buffer pH 4.5 using the USP II paddle apparatus.

Drug absorption was measured by UV. Stability studies were conducted on tablets placed in accelerated storage conditions at 40°C/75% RH in closed containers.

Figure 1: Direct compression followed by curing at 60°C during 30 min.



4 - RESULTS AND DISCUSSION

Glycerides are non-ionic; hence the lipid matrix is not impacted by pH changes. However diffusion rate from the matrix can be influenced by drug solubility. For Theophylline, there is no difference of drug release when tablet is exposed to various pHs (Figure 2).

Glycerides are non soluble in ethanol and therefore the lipid matrix is not subject to alcohol dose dumping. For Theophylline, being more soluble in ethanol than in water, it is likely that the diffusion rate will be faster from a tablet. Then it is important to apply a hot melt treatment to fuse the drug with lipids that reduces its surface of contact with ethanolic media. The simple curing at 60°C for 30 minutes generates the melt of Precirol® ATO 5 in the tablet and confers a cement-like network that makes Theophylline diffusion independent from alcohol content (Figure 3).

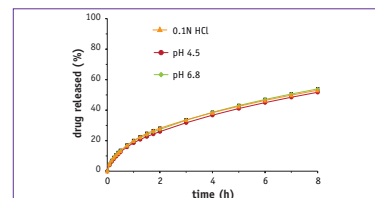


Figure 2: Effect of pH on theophylline release from Compritol® 888: Precirol® ATO 5 matrix tablets.

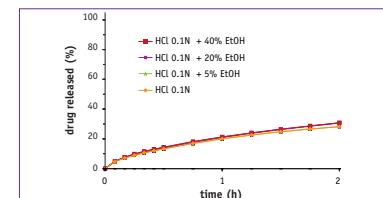


Figure 3: Influence of ethanol concentrations on theophylline release from Compritol® 888 ATO: Precirol® ATO 5 matrix tablets.

Interestingly, pictures showing the angle of contact of water on the tablet surface demonstrate that the curing treatment at 60°C increases the lipophilicity of the tablet surface.

The angle of contact was increased from 86° before curing to 125° after curing. Figure 4 shows how the droplet of water wets the tablet surface during 90 seconds. For a non-cured tablet, water droplet collapses slowly because it starts to dissolve hydrophilic ingredients on the surface.

For the cured tablet, Precirol® has created a lipid film on the surface so that the water droplet remains spherical during the 90 seconds. The presence of this film confirms that Precirol® ATO 5 melts during the curing step and confers a cement-like network within the tablet, which renders the surface lipophilic.

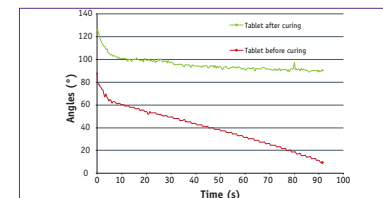
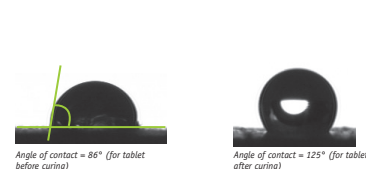


Figure 4: Variation of the contact angle function of time (s.).

To study the long term stability of the lipid matrix, tablets were placed into an oven at 25°C/60% RH and 40°C/75% RH. Figures 5 and 6 show that drug release is stable for 5 months. The study is still ongoing.

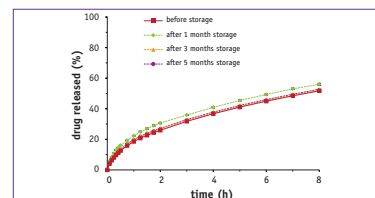


Figure 5: Dissolution of tablets in phosphate buffer pH 4.5. Stability of Compritol® 888: Precirol® ATO 5 matrix tablets upon long term storage at 25°C/60% RH.

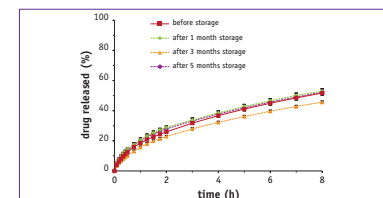


Figure 6: Dissolution of tablets in phosphate buffer pH 4.5. Stability of Compritol® 888: Precirol® ATO 5 matrix tablets upon long term storage at 40°C/75% RH.

5 - CONCLUSION

The combination of Precirol® ATO 5 and Compritol® 888 ATO in lipid matrix tablets can offer a great opportunity to increase the tablet stability and robustness for drug subject to alcohol dose dumping. The process is straightforward: direct compression followed by a short curing step at 60°C for 30 min.